Full Stack & DevOps Engineer

AWS Cloud, CI/CD, Generative AI & MCP

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SUMMARY

Full Stack & DevOps engineer with 5 years of experience on AWS. I design and deliver complete applications, from Python and Node.js backends to Angular front ends, with their Terraform and Kubernetes infrastructure, GitLab CI pipelines and tests, from TDD to end-to-end. I bring generative AI into production (AI agents, MCP servers) and single-handedly delivered an LLM gateway of 16 microservices.

TECHNICAL SKILLS

Cloud & DevOps	AWS (Lambda, API Gateway, S3, SQS, Cognito, EventBridge, CloudFront, IAM, CloudWatch) · Terraform · Terragrunt · Docker · Kubernetes · ArgoCD · GitLab CI · HashiCorp Vault
Generative AI	LLMs · LLMOps · AI governance (human validation, cost caps) · AI agents (LangGraph) · Model Context Protocol (MCP) · Vector search · Langfuse · Apache Airflow
Languages & Frameworks	Python (FastAPI, Chalice) · TypeScript · JavaScript · Node.js (Express) · Angular · React / Next.js
Databases	MongoDB · MySQL · SQL / NoSQL
Testing & Quality	TDD · Automated unit and integration tests in CI/CD · Blocking coverage gates · End-to-end tests with Playwright via Playwright MCP · pytest · Vitest
Methods & Tools	Agile (Scrum) · Git · Microservices · Serverless and event-driven architectures

WORK EXPERIENCE

Full Stack & DevOps Engineer, Portago

12/2024 - Present

LLM Gateway - Enterprise AI gateway

Context: OpenAI-compatible API that routes every call from the company's products to LLM providers.

- Independently designed and took to production an LLM gateway of 16 **AWS Lambda** microservices.
- One OpenAI-compatible API for chat, embeddings and audio: products integrate with no code changes.
- Service continuity: automatic LLM provider failover, retries on errors and SSE streaming.
- Capped AI spending per product: budget reserved before every call, alerts and FinOps reporting.
- Security: provider keys in **Vault**, multi-tenant isolation, SSRF protection and an audit log.
- AI tool governance: **MCP** servers ranked by risk level and **Langfuse** tracing of every request.
- Easier integration for product teams: a **Python** client library and API contract tests.
- End-to-end tests of the **Angular** console with **Playwright**, driven through **Playwright MCP**.
- **Terraform/Terragrunt** infrastructure across 3 environments, deployed through **GitLab CI** pipelines.

Technologies: Python · Chalice · AWS Lambda · API Gateway · Cognito · Vault · Langfuse · MCP

Moveo - HR skills and internal mobility platform

Context: One application for skills, training, internal mobility and recruitment, with AI supporting employees and HR.

- Developed 2 AI assistants, for employees and HR: a **LangGraph** agent and a 132-tool **MCP** server.
- Transparent AI for users: the agent's steps shown live and model selection in the chat.
- Designed multi-company serverless microservices: skills, positions, quizzes and training.
- Built asynchronous AI generation via **SQS** of learning paths, quizzes, CV analyses and career advice.
- Human-controlled AI: two-step confirmation and HR validation of expert skill levels.
- Developed the 3 **Angular** areas, employee, HR and admin, each limited to its role's permissions.
- Containerised the agent services on **Kubernetes**, with state control and sync by **ArgoCD**.

Technologies: Python · FastAPI · Generative AI · LangGraph · MCP · Kubernetes · ArgoCD · SQS

Full Stack & DevOps Developer, Allence Cloud

11/2021 - 11/2024 | Tunisia

Career - AI-assisted recruitment platform

Context: Automated candidate pre-screening: CV analysis, scoring and AI-led interviews.

- Automated pre-screening: offers in 4 formats, LLM-based CV analysis and a score out of 100.
- AI interviews in 3 formats: written, voice and phone with **Twilio** and **OpenAI Realtime**.
- Built candidate-to-offer matching with **MongoDB** vector search over a 530-skill catalogue.
- Recruiter AI agent on **LangGraph**: 60-tool **MCP** server and history compression.
- Candidate assessment in the chat: quizzes generated by the AI agent and graded in the conversation.
- Decision support for recruiters: side-by-side candidate comparison and detailed evaluation reports.
- Event-driven **SQS** and **Lambda** workers, plus containerised **FastAPI** microservices on **Kubernetes**.
- Reliable releases: test-driven development and a blocking coverage gate in CI.

Technologies: Python · FastAPI · Generative AI · LangGraph · MCP · Twilio · OpenAI Realtime

PhotoEvent - Event photography marketplace

Context: Event photo sales, with photo search by face recognition.

- Designed a serverless backend of 17 **Lambda** functions and 104 API routes, with an **Angular** front end.
- Photo search by face recognition with **Amazon Rekognition**, one collection per event.
- Automated gallery import with an **Apache Airflow** DAG on **Kubernetes**.
- Reliable photographer finances: orders, commissions and payouts in **MongoDB** transactions.
- Generative AI moderation of reviews, and image protection with watermarks and signed links.
- Built photographer onboarding: legal document upload and administrator approval.
- Set up customer support: prioritised tickets, status tracking and email notifications via **SQS**.
- End-to-end tests of the **Angular** front end with **Playwright MCP**, alongside pytest suites on the backend.
- Independent releases per domain: one repository and one pipeline per service, 80 % minimum coverage.

Technologies: Python · Chalice · Generative AI · Rekognition · Airflow · S3 · SQS · Terraform · Playwright

Full Stack Developer, Dida Mind

02/2021 - 07/2021 | Tunisia

ProHunting - Hunting tourism platform

Context: Booking and management of hunting stays in Tunisia for European clients.

- Designed and built the complete application: **Node.js/Express** API and **Next.js** front end.
- Built the back office that runs the full stay lifecycle, from request to closing.
- Regulatory compliance: hunting permits tracked by zone and by day, official fields checked at entry.
- Integrated a partner hotel-booking API and real-time team notifications with **Socket.IO**.
- Launched the public website in French, English and German, with content and SEO managed by the team.
- Automated stay documents: templates with merge fields and PDF generation with **Puppeteer**.
- Secured the back office: role- and user-level permissions, account deletion restricted to super-admins.

Technologies: Node.js · Express · Next.js · TypeScript · MySQL · Socket.IO · Puppeteer · Playwright

EDUCATION

- Engineering Degree in Software Engineering - ESPRIT, Tunisia, 2024
- Bachelor's Degree in Computer Science - ISTIC Borj Cedria, Tunisia, 2021

CERTIFICATIONS AND LANGUAGES

- AWS Certified Developer – Associate - AWS, 2026
- Docker Certified Associate (DCA) - Docker, 2026
- Astronomer Certified for AI Orchestration Fundamentals - Astronomer, 2026
- Model Context Protocol: Advanced Topics - Anthropic, 2026
- Introduction to Model Context Protocol - Anthropic, 2026
- Multicloud Network Associate - Aviatrix, 2025

Languages: French (fluent) | English (fluent) | Arabic (native)